SQL Server 2019 CPT2.4

Gianluca Hotz

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Who am I?

- Gianluca Hotz | @glhotz | ghotz@ugiss.org
- Independent Consultant, Founder and Mentor SolidQ
  - 20+ years on SQL Server (from 4.21 in 1996)
  - Database modeling and development, sizing and administration, upgrade and migration, performance tuning
- Interests
  - Relational model, DBMS architecture, Security, High Availability and Disaster Recovery
- Community
  - 20 years Microsoft MVP SQL Server (from 1998)
  - Founder and President UGISS
    - User Group Italiano SQL Server (PASS Chapter)

Configuration

Installation

- Most new things in 2016/2017
  - Separate downloads
  - IFI and tempdb configuration
  - Linux
  - R and Python integration
  - Integration Services Scale-out
  - Polybase

PolyBase Java Connector for HDFS

SQL Server 2019 CTP2.2 Setup

Feature Selection

Select the Developer features to install.

- Product Key
- License Terms
- Global Rules
- Microsoft Update
- Product Updates
- Install Setup Files
- Install Rules
- **Feature Selection**
- Feature Rules
- Instance Configuration
- PolyBase Configuration
- Server Configuration
- Database Engine Configuration
- Consent to install Microsoft R ...
- Consent to install Python
- Feature Configuration Rules
- Ready to Install
- Installation Progress
- Complete

Looking for Reporting Services? Download it from the web

Features:

- Instance Features
  - [x] Database Engine Services
    - [x] SQL Server Replication
    - [x] Machine Learning Services (In-Database)
      - [x] R
      - [x] Python
    - [ ] Full-Text and Semantic Extractions for Search
    - [ ] Data Quality Services
    - [x] PolyBase Query Service for External Data
      - [x] Java connector for HDFS data sources
  - [ ] Analysis Services
- Shared Features
  - [ ] Machine Learning Server (Standalone)
    - [ ] R
    - [ ] Python
  - [ ] Data Quality Client

Feature description:

Includes PolyBase Java Connector that enables truly integrated querying across HDFS data using standard T-SQL statements.

Prerequisites for selected features:

- Microsoft MPI v9

Needs to be manually installed:

- Microsoft Visual C++ 2012 Redistributable
- Oracle SE Java Runtime Environment Version 7 Update 51 or higher (64-bit)

Disk Space Requirements

Drive C: 11509 MB required, 116830 MB available

Select All | Unselect All

Instance root directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory: C:\Program Files\Microsoft SQL Server\

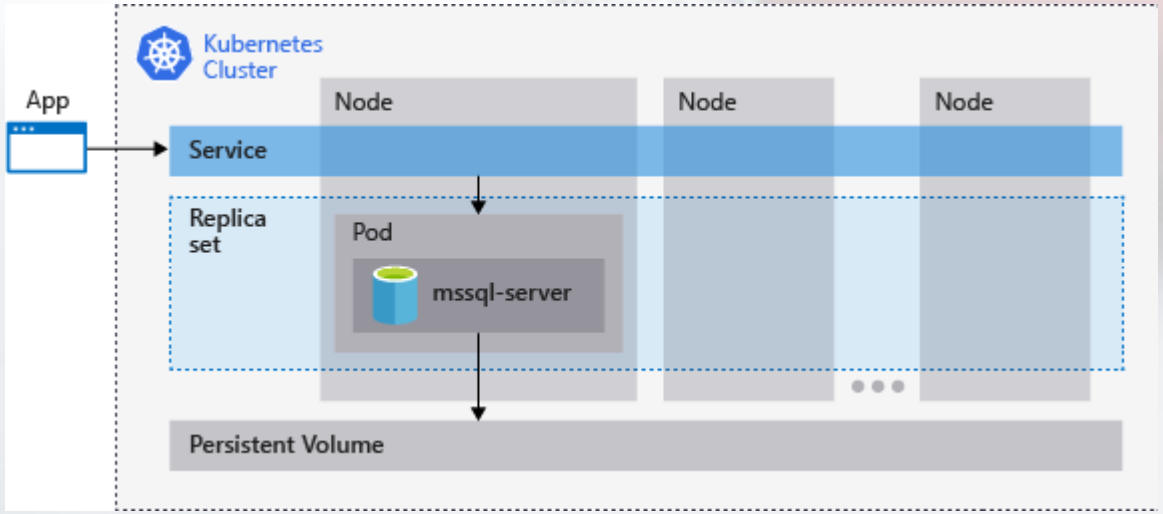
Shared feature directory (x86): C:\Program Files (x86)\Microsoft SQL Server\

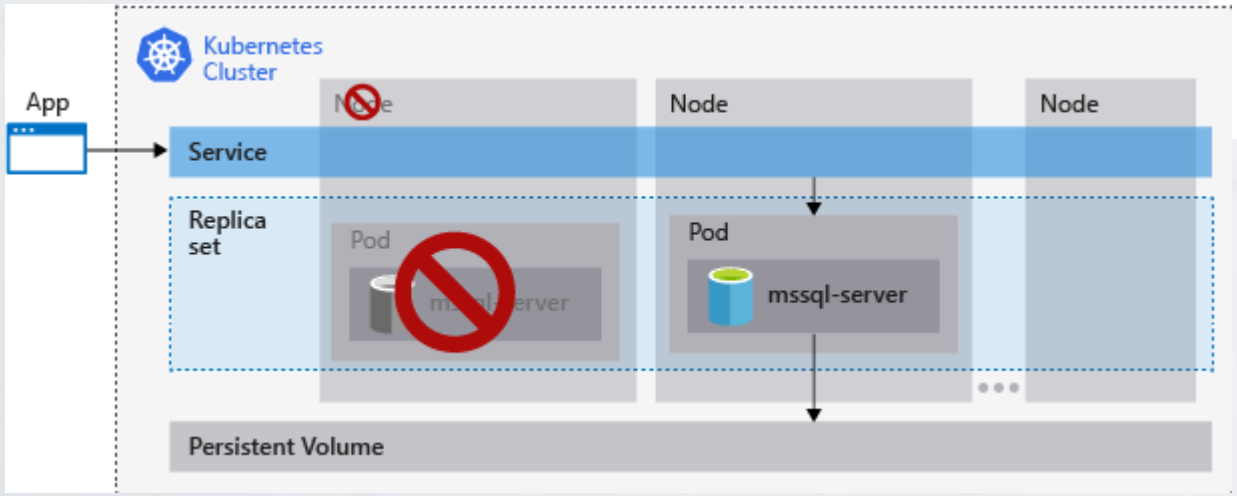
< Back | Next > | Cancel

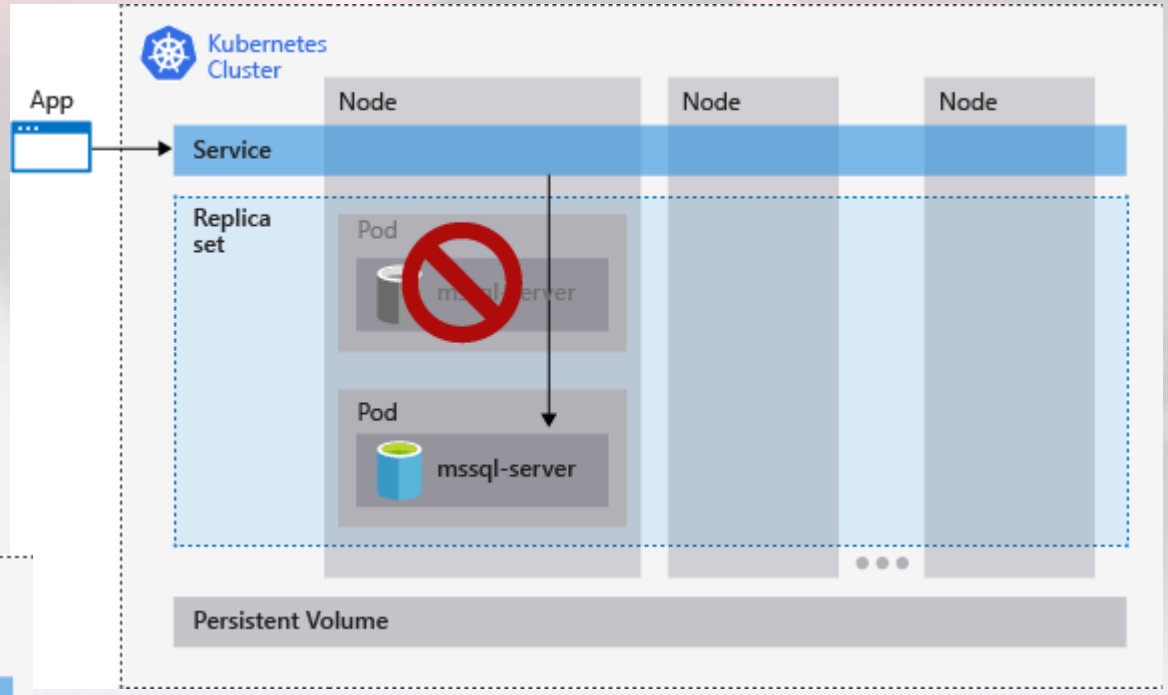
SQL Server on Linux

- Replication support
  - Snapshot, Transactional and Merge
- Support for Microsoft Distributed Transaction Coordinator (MSDTC)
- Always On Availability Group on Docker containers with Kubernetes
  - Kubernetes operator deploys StatefulSet including container with mssql-server container and health monitor
- OpenLDAP support for third-party AD providers
- Machine Learning Services (In-Database) on Linux
- New container registry

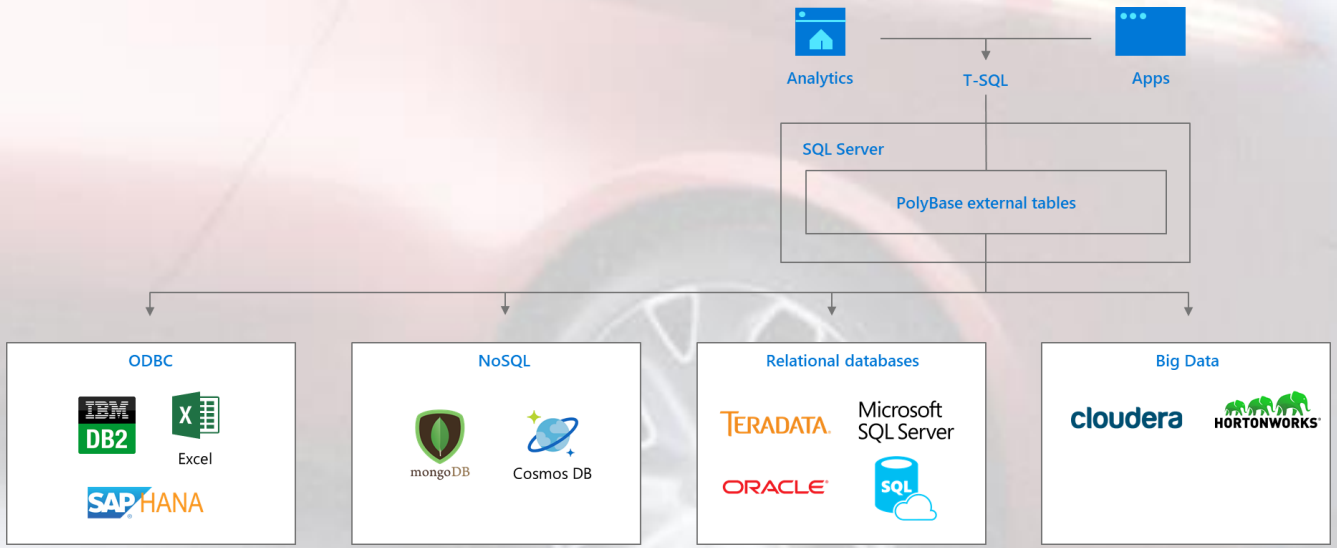
HA solution in Azure Kubernetes Service







Polybase



- Data Virtualization
  - External Data Source
  - External Tables
- SQL Server 2016+
  - Azure Blob Storage, Hadoop
- SQL Server 2019
  - SQL Server (all flavors), Oracle, Teradata, MongoDB, CosmosDB (Mongo API)
  - ODBC Generic Type (e.g. DB2, SAP Hana, Excel, …)

External Tables vs Linked Servers

External Tables

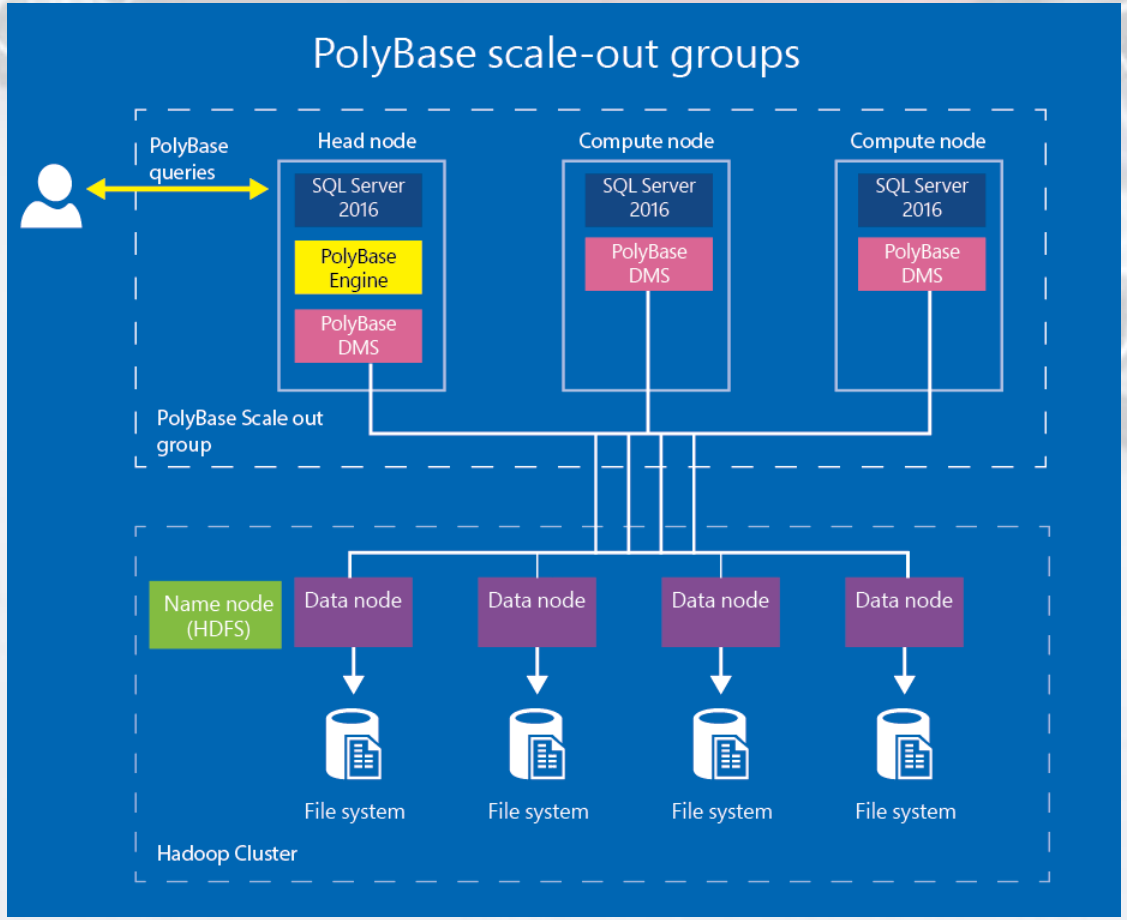
- Database Scoped
- ODBC Drivers
- Read-only*
- Scale out queries with push-down
- Failover with AG
- Basic authentication
- Distributed Transactions not supported

Linked Servers

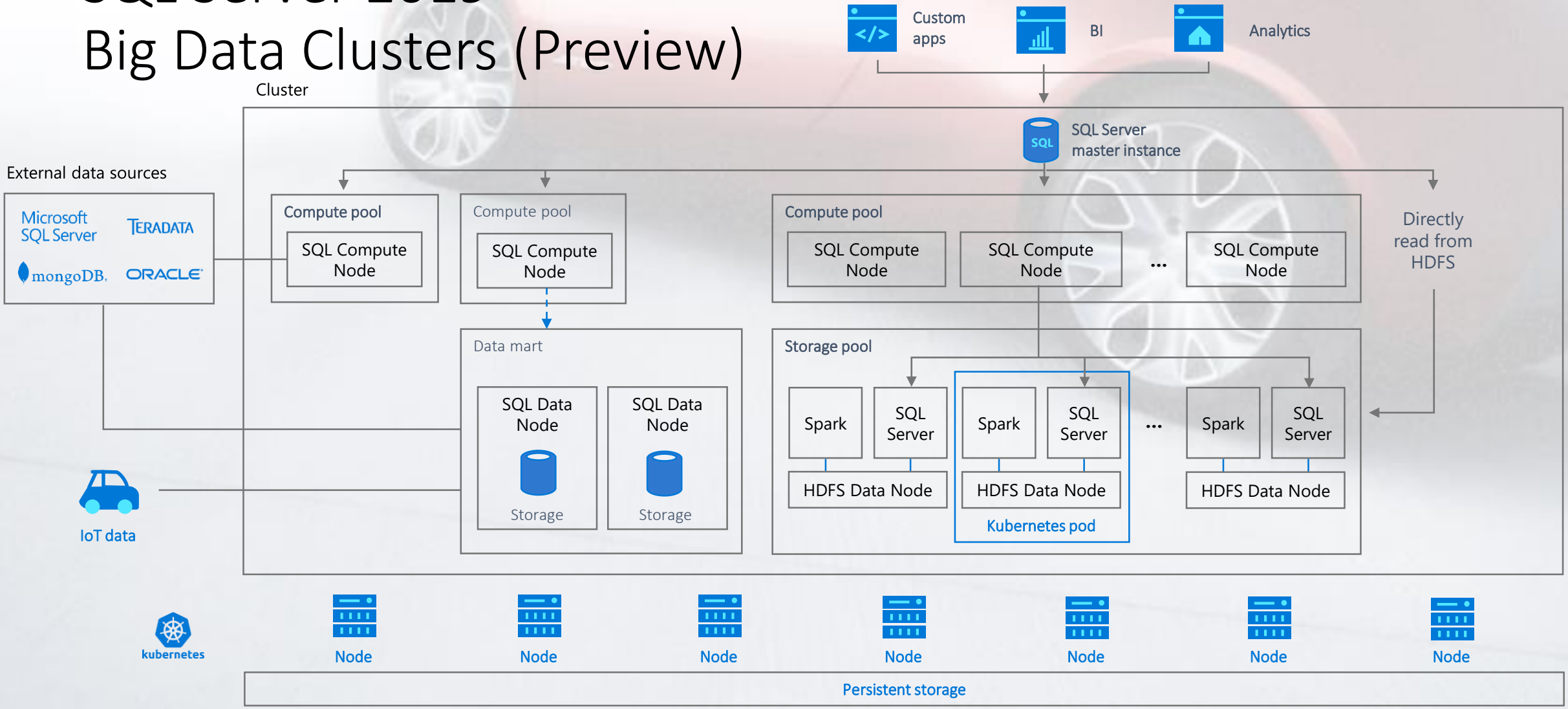
- Instance Scoped
- OLEDB Providers
- Read/write
- Single threaded queries with push-down
- Requires separate config from AG
- Basic and integrated authentication
- Distributed Transactions supported

* Insert into HDFS allowed

Polybase Scale-Out Groups



SQL Server 2019
Big Data Clusters (Preview)



Polybase in Big Data Cluster vs Stand-alone

Feature	Big data cluster	Stand alone
Create external data source for SQL Server, Oracle, Teradata, and Mongo DB	X	X
Create external data source using a compatible third-party ODBC Driver		X
Create external data source for HADOOP data source	X	X
Create external data source for Azure Blob Storage	X	X
Create external table on a SQL Server data pool	X	
Create external table on a SQL Server storage pool	X	
Scale-out query execution	X	X

Other Services

- Master Data Services
  - Silverlight controls replaced with HTML
- SQL Server Machine Learning Services
  - Windows Server Failover Cluster support
  - Partition-Based modeling

Administration

Tools

- SQL Server Management Studio V18 (Release Candidate 1)
- Azure Data Studio (was Operations Studio)

Resumable online index operations

- Resume after failure (e.g. out of disk space)
- Pause and resume later (e.g. free temporarily resources)
- Manage large indexes using less log an shorter transactions
- Fit rebuild operations into limited maintenance windows
  - REBUILD
    WITH (ONLINE = ON, RESUMABLE = ON, **MAX_DURATION = 30 MINUTES**);
- SQL Server 2017+ Resumable Index Rebuild
- SQL Server 2019+ Resumable Index Create

Transparent Data Encryption

- ALTER DATABASE … SET ENCRYPTION **SUSPEND**
- ALTER DATABASE … SET ENCRYPTION **RESUME**

Columnstore Indexes

- **Online** build/rebuild Clustered Columnstore
  - E.g. conversion from row-store to Columnstore

DBCC CLONEDATABASE

- Instantaneous schema-only copy of a database for troubleshooting
  - No data, full-schema, statistics and Query Store
  - Non-blocking
  - Read-only by default (can be changed)
  - Optionally **NO_STATISTICS**, **NO_QUERYSTORE**
  - SQL Server 2012 SP4, 2014 SP2 CU3, 2016 SP1, 2017
- New in SQL Server 2019
  - Columnstore Statistics

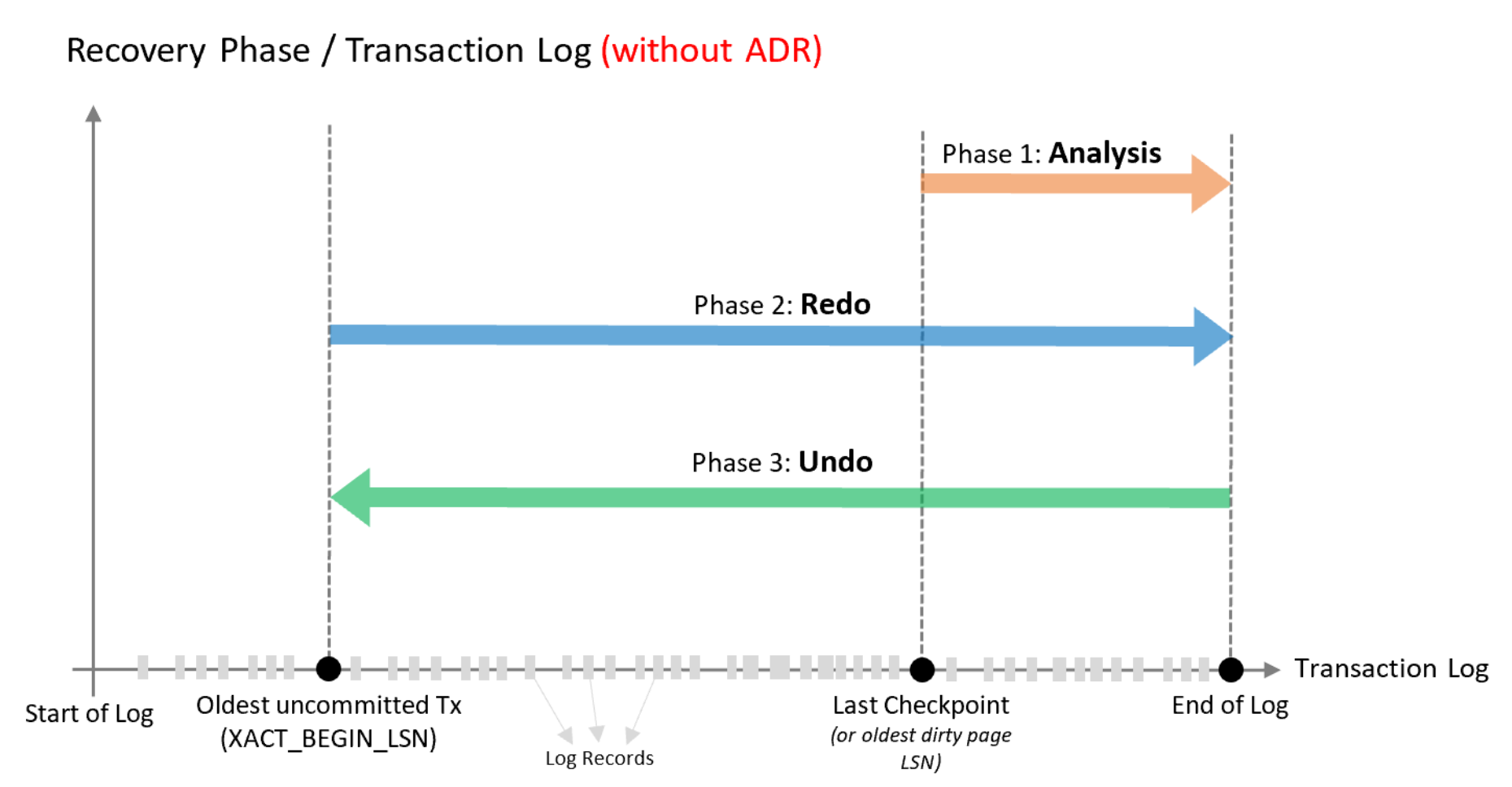
Accelerated Database Recovery

- Benefits
  - Fast and consistent database recovery
    - Number/size of active transactions don't impact recovery time
  - Instantaneous transaction rollback
    - Active time and number of updates don't impact rollback time
  - Aggressive Log Truncation
    - Even with long running transactions, prevents growing out of control
- High level
  - Versioning all physical database modifications
  - Only logical operations undone (are limited and can be undone instantly)
  - Active transactions at crash time are marked as aborted
  - Any versions generated aborted transactions can be ignored user queries
- Currently available in Preview in Azure SQL Database and CTP2.3+!

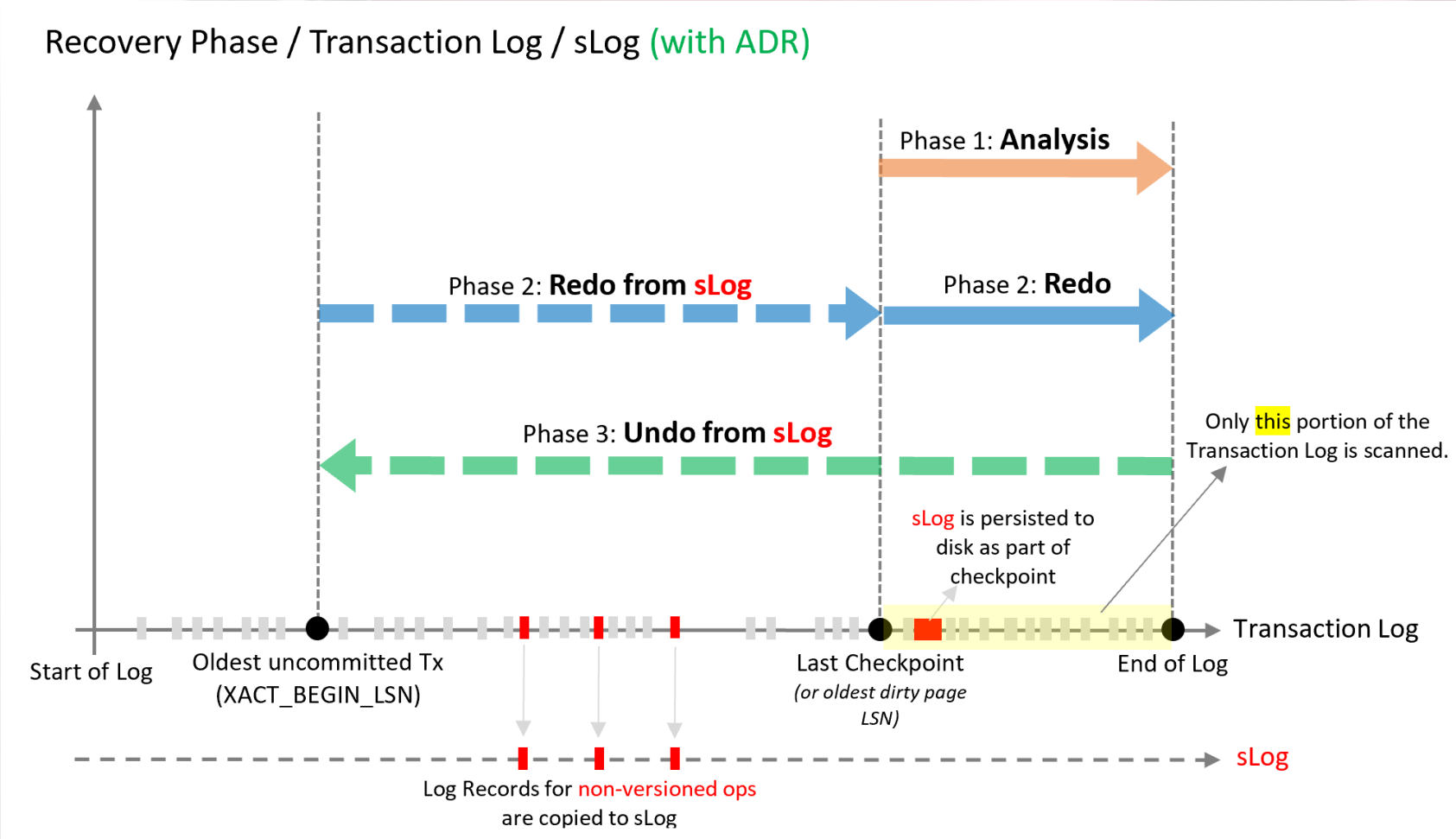
Accelerated Database Recovery Components

- Persisted Version Store (PVS)
  - New version store, stored in the database instead of tempdb
  - Enable also resource isolation
- Logical Revert
  - Asynchronous process performing row level version based undo
    - Keeps track of all aborted transactions
    - Performs rollback using PVS
    - Releases all locks immediately after transaction abort
- sLog
  - Secondary log stream storing log records for non versioned operations
    - Low volume and in-memory
    - Serialized on disk during CHECKPOINT
    - Enables aggressive transaction log truncation
- Cleaner
  - Asynchronous process that cleans page versions

Current database recovery process



Accelerated database recovery process



Improved diagnostic data for stats blocking

- Query Waiting for synchronous update operations
- Now **sys.dm_exec_requests** shows **SELECT (STATMAN)**
- New **WAIT_ON_SYNC_STATISTICS_REFRESH** wait stat

session_id	request_id	start_time	status	command
59	0	2018-11-13 11:50:36.617	suspended	SELECT (STATMAN)

wait_type	waiting_tasks_count	wait_time_ms	max_wait_time_ms	signal_wait_time_ms
WAIT_ON_SYNC_STATISTICS_REFRESH	1	18781	18781	0

Availability Groups Enhancements

- More synchronous replicas
  - SQL Server 2012 4 replicas, 2 synchronous
  - SQL Server 2014 8 replicas, 2 synchronous
  - SQL Server 2017 8 replicas, 3 synchronous
  - SQL Server 2019 8 replicas, **5 synchronous**
- Secondary-to-primary read/write redirection
  - READ_WRITE_ROUTING_URL and ApplicationIntent=ReadWrite (default)
  - Killer feature to replace Listener
    - Cluster technology not offering listener-like features
    - Multi-subnet scenarios too complex to setup/maintain (e.g. Pacemaker)
    - Read scale-out or DR with cluster type NONE

Storage Class Memory / PMEM

- Allows low latency I/O
  - memory-mapped memcpy-like operations in user mode
- SQL Server 2016 SP1
  - NVDIMM-N for tail of the log caching
- SQL Server 2019
  - PMEM devices Linux
  - support for data, log In-Memory OLTP checkpoint files placement
- Hybrid Buffer Pool
  - Clean pages direct referenced on PMEM devices without copy
  - Dirty pages still kept in DRAM

Estimating Data Compression savings

- **sp_estimate_data_compression_savings**
  - Returns specified object's current size and estimates
- SQL Server < 2019
  - ROW and PAGE compression
- SQL Server 2019+
  - Adds **COLUMNSTORE** and **COLUMNSTORE_ARCHIVE** compression
  - Object type determines Columnstore type
    - E.g. Heap -> Clustered, Clustered index -> Clustered

Query Profiling Infrastructure

- Run-time information on query execution plans
  - E.g. live query statistics
- Lightweight profiling
  - introduced in SQL Server 2014 SP2, 2016 SP1+
  - 2% expected CPU overhead vs. 75%
  - SQL Server 2019+
    - Enabled by default
    - XEvents **query_post_execution_plan_profile** (vs **query_post_execution_showplan**)
    - DMF **sys.dm_exec_query_plan_stats()** last known ***actual*** execution plan

Internal database pages information

- Undocumented DBCC PAGE
- New in SQL Server 2019
  - **sys.dm_db_page_info(DatabaseId, FileId, PageId, Mode)**
  - **page_resource** column in **sys.dm_exec_requests** and **sys.sysprocesses**
  - **sys.fn_PageResCracker(page_resource)** to get **db_id, file_id, page_id**

Results | Messages

	database_id	file_id	page_id	page_header_version	page_type	page_type_desc	page_type_flag_bits	page_type_flag_bits_desc	page_flag_bits	page_flag_bits_desc
1	2	1	1	1	11	PFS_PAGE	0x0		0x0	
2	2	1	2	1	8	GAM_PAGE	0x0		0x0	
3	2	1	3	1	9	SGAM_PAGE	0x0		0x0	
4	2	1	4	NULL	NULL	NULL	NULL	NULL	NULL	NULL
5	2	1	5	NULL	NULL	NULL	NULL	NULL	NULL	NULL
6	2	1	6	1	16	DIFF_MAP_PAGE	0x0		0x200	HAS_CHECKSUM
7	2	1	7	1	17	ML_MAP_PAGE	0x0		0x0	
8	2	1	8	1	1	DATA_PAGE	0x4		0x200	HAS_CHECKSUM
9	2	1	9	1	13	BOOT_PAGE	0x0		0x0	
10	2	1	10	1	10	IAM_PAGE	0x0		0x200	HAS_CHECKSUM
11	2	1	11	1	2	INDEX_PAGE	0x4		0x200	HAS_CHECKSUM

Demo

- Compression Savings
- Internal Pages Information

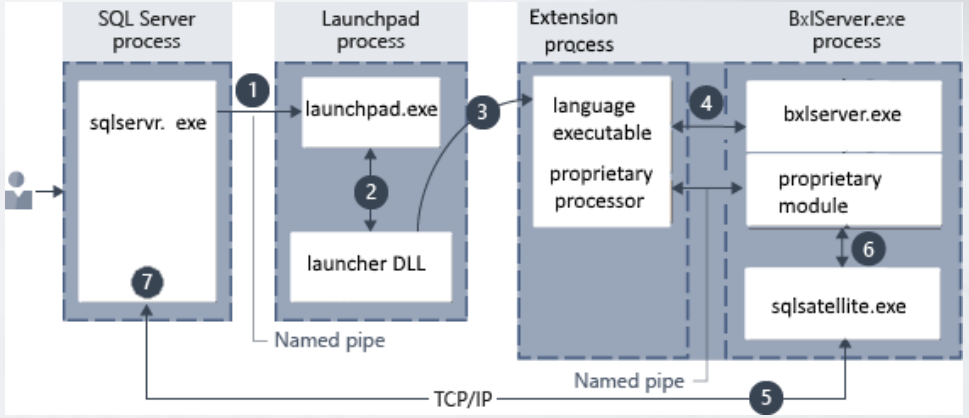
Development

Truncation error messages

- Error message 8152 too generic
  - String or binary data would be truncated
- SQL Server 2019 introduces message 2628
  - String or binary data would be truncated in **table** '%.*ls', **column** '%.*ls'. Truncated **value**: '%.*ls’
- Enabled with Trace Flag 460
  - Opt-in required to avoid breaking parsing applications

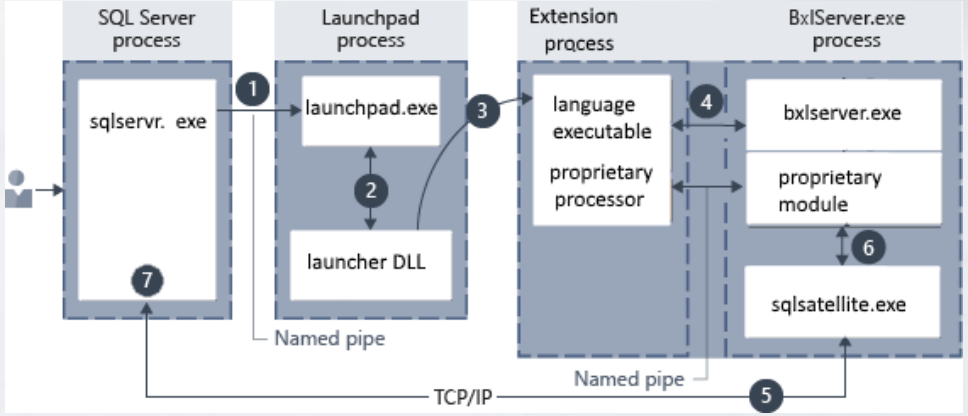
Extensibility Framework

- Secure external script execution
- Scale/optimization opportunities
- SQL Server integration (e.g. store procedures, PREDICT)
- Language Support
  - SQL Server 2016+ Support for R
  - SQL Server 2017+ Support for Python
  - SQL Server 2019+ Support for Java



Extensibility Framework Components

- Launchpad Service
  - One per SQL Server instance (with Machine Learning Services)
  - Provides security isolation
- BxlServer
  - Manage communications between SQL Server and external processes
  - Binary Exchange Language data format
- SQLSatellite
  - Extensibility API used by BxlServer
  - I/O data/arguments , error handling



Java Language extension

- Leverages Extensibility Framework
  - Through **sp_execute_external_script**
- Current support
  - On Windows version 1.10 (JRE 10, JDK 10)
  - On Linux version 1.8 (JRE 8, JDK 8)

UTF-8 Support

- Full support for import/export, collations, replication, …
  - Still not for Linked Servers, In-Memory OLTP, External Table (Polybase)
- CHAR and VARCHAR support (Windows collations only)
- UTF8 in collation names
  - E.g. LATIN1_GENERAL_100_CI_AS_SC_UTF8
- Can provide storage savings
  - E.g. up to 50% from NCHAR(10) to CHAR(10) with UTF8 vs UTF16

SQL Graph enhancements

- Derived tables and view support in MATCH queries
  - Set of nodes/edges using UNION ALL
  - Useful for heterogeneous entities or connections between them
- MATCH support in MERGE
- Edge Constraints
  - CONNECTION constraint

Demo

- Java integration

Security

Certificate Management

- Extended in SQL Server Configuration manager
  - View and validate certificates installed
  - View certificates close to expiration
  - Deploy Certificates across machine in Availability Groups
  - Deploy Certificates across machine in Failover Cluster Instances

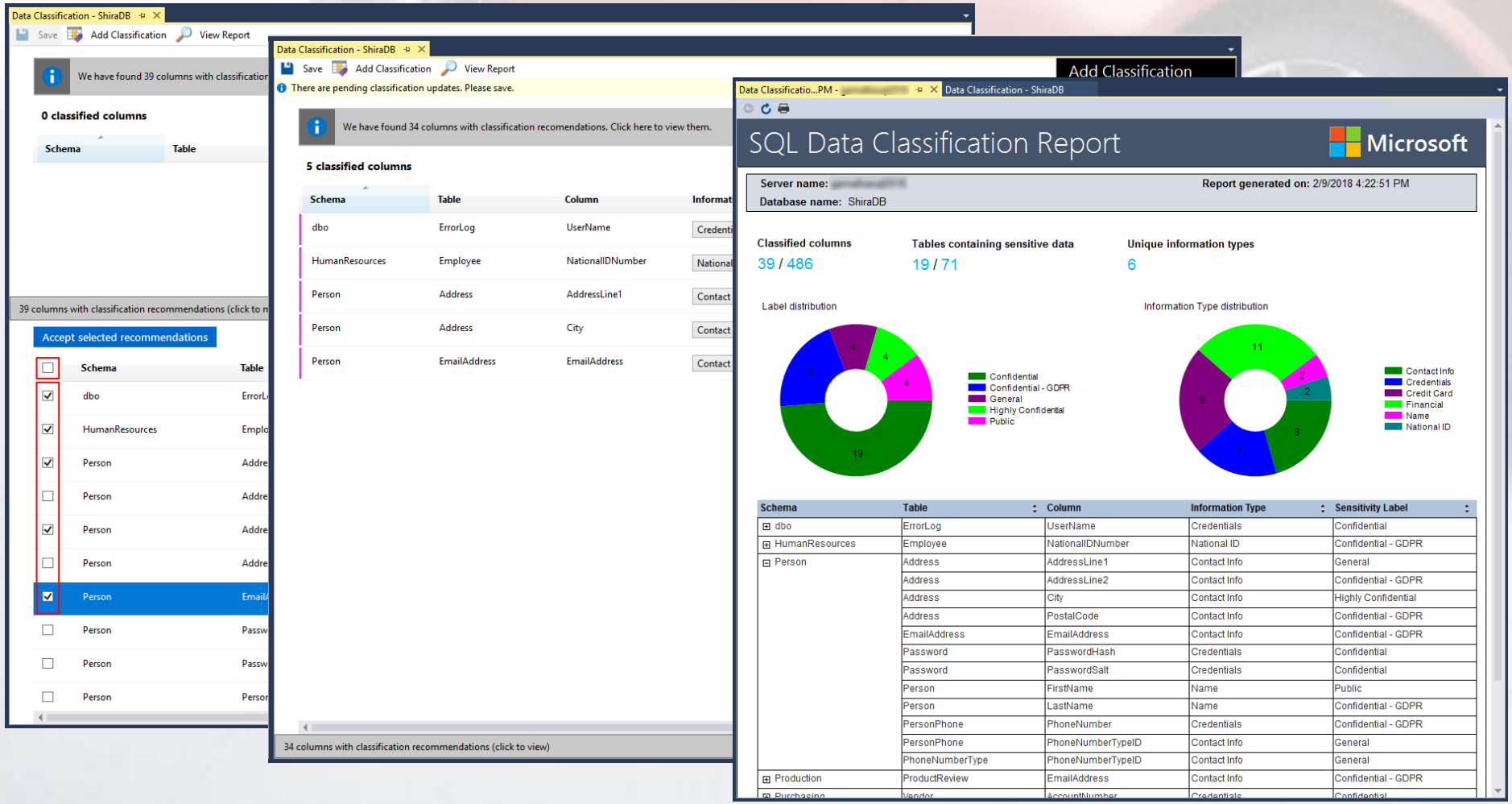
SQL Data Discovery and Classification

- SQL Server management Studio Tool (V17.5)
- Discovery & Recommendations, Labeling, Reporting
- Metadata con be persisted and queried
  - Based on Extended Properties
    - sys_information_type_name, sys_sensitivity_label_name
- Support for SQL Server 2008+ and Azure SQL Database

SQL Server Sensitivity Classification

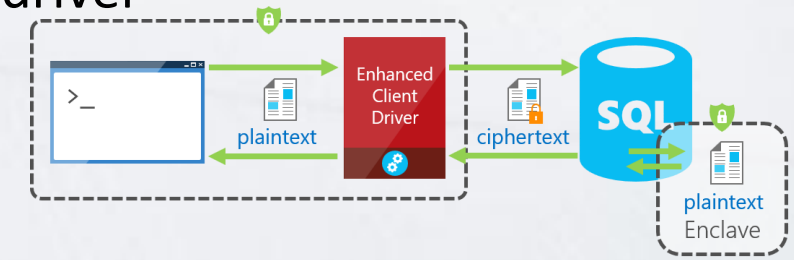
- SQL Server 2019+
- (already available in Azure SQL Database)
- T-SQL command **ADD|DROP SENSITIVITY CLASSIFICATION**
  - applies to tables, columns
  - **LABEL**, **LAtBEL_ID**, **INFORMATION_TYPE**, **INFORMATION_TYPE_ID**
- Metadata stored in **sys.sensitivity_classifications**
- SQL Server Audit add column **data_sensitivity_information**

Data Discovery and Classification Demo



Always Encrypted with Secure Enclaves

- Basic architecture as SQL Server 2016+ implementation
- Now allows server-side computation on encrypted columns
  - In-Place Encryption (ALTER TABLE for initial encryption)
  - Rich computations (e.g. range comparisons, LIKE predicates, …)
- Inside secure enclaves
  - Virtualization-based Security (VBS) secure memory enclaves
    - also known as Virtual Secure Mode(VSM) enclaves
  - Operation on plaintexts cannot be disclosed outside enclave
  - Column Master Keys sent over secure channel by client driver
- Still some limitations (no indexing)
- Performance optimizations pending…



Data Masking

Dynamic Data Masking

- On the original database
- Original data intact
- On-the-fly at query time
- Based on user permissions

Static Data Masking

- On a copy of the database
- Original data not retrievable
- At storage level
- Masked for everyone

Static Data Masking

- Component of SQL Server Management Studio V18 Preview5+
- Define per-column masking configuration
  - NULL, Single-Value, Shuffle, Group Shuffle, String Composite
  - Can save and load it
- It’s basically a backup/restore and modify data according to config
- No automation yet ☹

Static Data Masking Limitations

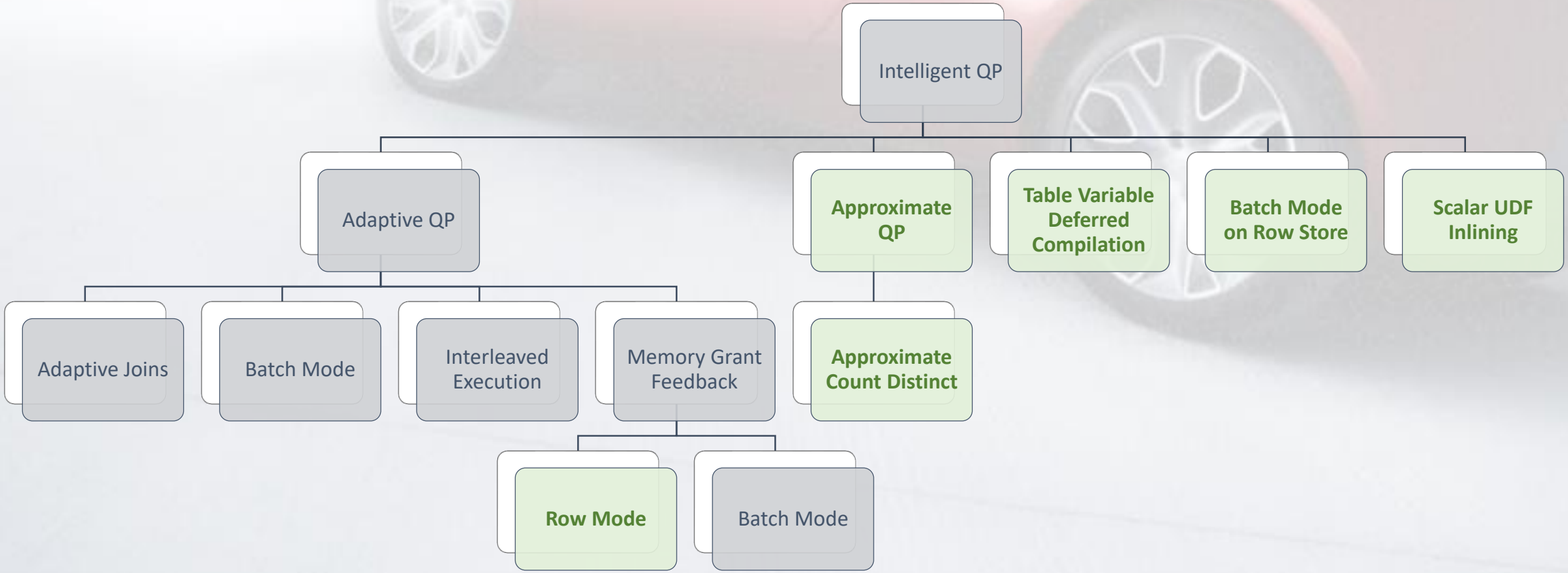
- No temporal and memory-optimzed tables
- No computed and identity columns
- No geometry and geography types
- Azure SQL Database Hyperscale service tier not supported
- Statistics not updated
- No cleanup in case of error
  - can leave sensitive data copies (backupset)
- Data and log files may contain sensitive data
  - retrievable with hex editor

Demo

- Data Classification

Performance

Intelligent Query Processing



Execution Modes

- Row Mode
  - Execution tree iterators consume 1 row at a time
  - Traditional execution mode for Rowstore
- Batch Mode
  - Execution tree iterators consume a batch of rows at a time
  - Optimal with large scan operations (e.g. large table aggregates or joins)
  - SQL Server 2012 introduced to leverage Columnstore Indexes
  - SQL Server 2016/2017 extended usage scenarios for CI
  - SQL Server 2019 extended usage scenarios to Rowstore

Batch Mode on RowStore

- Help reducing CPU Consumption
- Columnstore still a better choice
  - for OLAP workload that is I/O bound
  - can’t always create it (e.g. impact on OLTP, features not supported)
- Limitations
  - In-Memory tables not supported (only heaps & disk-based b-trees)
  - Not used when fetching/filtering LOB columns
  - (including sparse columns sets & XML)

Batch Mode on Rowstore Control

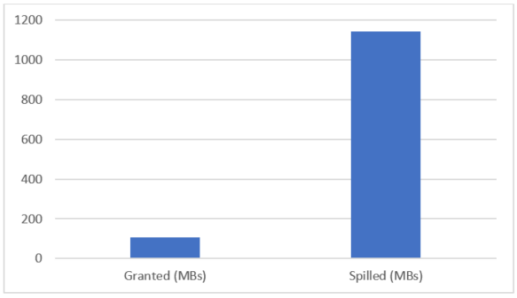
- SQL Server < 2019
  - Some scenarios covered with tricks… (article part1, part2, part3)
- SQL Server 2019+
  - Scenarios supported directly by Query Processor
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **BATCH_MODE_ON_ROWSTORE** = ON|OFF
  - OPTION (USE HINT ('**ALLOW_BATCH_MODE**’));
  - OPTION (USE HINT (‘**DISALLOW_BATCH_MODE**'));

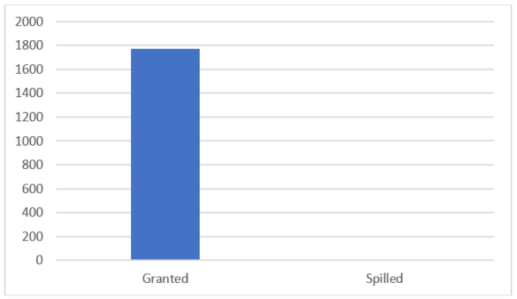
Memory Grant

- Excessive Grant
  - Too much memory allocated vs. memory used
  - Impact: blocking, out-of-memory, reduced concurrency
- Poor Grant
  - Not enough memory allocated resulting in data spill to tempdb
  - Impact: slow query, excessive disk usage (tempdb)
- Grant increase
  - dynamic grants increase allocation too much
  - impact: server instability, unpredictable performance

Memory Grant Feedback

- Post-execution evaluation
  - Updates grant value for cached plan
  - E.g. more memory if spilled, less if excessive grant
- Version support
  - SQL Server 2017+ Batch Mode
  - SQL Server 2019+ Row Mode
- Plan caching
  - Not persistent (i.e. not save in Query Store)
  - OPTION(RECOMPILE) prevents caching and memory grant feedback





Memory Grant Feedback Control

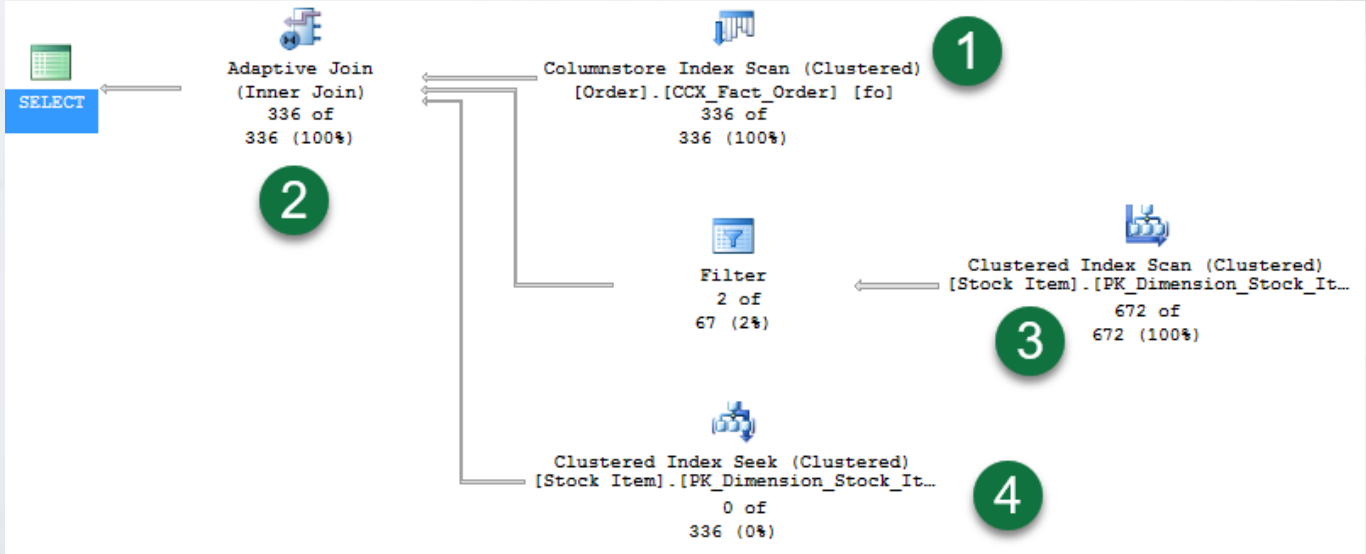
- Batch Mode
  - On by default with database compatibility level **140+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **BATCH_MODE_MEMORY_GRANT_FEEDBACK** = ON|OFF
  - OPTION (USE HINT('**DISABLE_BATCH_MODE_MEMORY_GRANT_FEEDBACK**'));
- Row Mode
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **ROW_MODE_MEMORY_GRANT_FEEDBACK** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_ROW_MODE_MEMORY_GRANT_FEEDBACK**'));

Troubleshooting Memory Grant Feedback

- Parameter sensitive scenarios
  - Some queries requires different plans with different grants
  - Memory grant feedback will disable itself when unstable
- Extended Events to monitor changes
  - SQL Server 2017+ **memory_grant_feedback_loop_disabled**
  - SQL Server 2019+ **memory_grant_updated_by_feedback**
- SQL Server 2019+ execution plan attributes
  - **IsMemoryGrantFeedbackAdjusted**
    - No: First Execution, Accurate Grant, Feedback disabled
    - Yes: Adjusting, Stable
  - **LastRequestedMemory**

Batch mode adaptive joins

- Scenario
  - Nested loop algorithm better for small build join inputs
  - Hash algorithm better for bigger inputs
- Adaptive joins defer choice after first input scanned



Interleaved Execution

- Problem with multi-statement table valued functions (MSTVFs)
  - SQL Server <= 2012 optimize with cardinality = 1
  - SQL Server 2014 & 2016 optimize with cardinality = 100
- SQL Server >= 2017
  - Start optimization
  - Pause and executes MSTVFs if candidate
  - Resume optimization with correct cardinality

Table Variable vs Temporary Tables

Area	Temporary Tables	Table Variables
Manual statistics creation and update	Yes	No\
Indexes	Yes	Only inline index definitions allowed
Constraints	Yes	Only PRIMARY KEY, UNIQUE and CHECK
Automatic statistics creation	Yes	No
Creating and using a temporary object in a single batch	Compilation of a statement that references a temporary table that doesn’t exist is deferred until the first execution of the statement	A statement that references a table variable is compiled along with all other statements before any statement that populates the Table Variable is executed, so compilation sees it as 1

Table Variable Deferred Compilation

- Before SQL Server 2019
  - Statement referencing TV compiled before population
  - Number of row estimate fixed at 1
- Starting with SQL Server 2019
  - Behaves like Temporary Tables
  - Statement referencing non existing TV is deferred until first execution
  - Number of row estimate much better
- Control
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION
    SET **DEFERRED_COMPILATION_TV** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_DEFERRED_COMPILATION_TV**'));

Scalar UDF inlining

- T-SQL user defined functions that returns a single data value
- Performance problems
  - Iterative invocation
    - once per row, context switching especially with query execution
  - Lack of costing
    - before, only relational operators were costed, assumption to be cheap…
  - Interpreted execution
    - each statement executes in isolation, no cross-statement optimizations
  - Serial execution
    - Intra-query parallelism not allowed

Scalar UDF Automatic inlining

- In SQL Server 2019 Scalar UDF automatically transformed into
  - Scalar Expressions
  - Scalar Subqueries
- Optimize the whole plan (UDFs no longer visible)
- Control
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION
    SET **TSQL_SCALAR_UDF_INLINING** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_TSQL_SCALAR_UDF_INLINING**'));
  - CREATE FUNCTION … WITH **INLINE** = ON | OFF

Scalar UDF inlining example

```
CREATE FUNCTION dbo.discount_price(@price DECIMAL(12,2), @discount DECIMAL(12,2))
RETURNS DECIMAL (12,2) AS BEGIN RETURN @price * (1 - @discount); END

SELECT L_SHIPDATE, O_SHIPPRIORITY
, SUM(dbo.discount_price(L_EXTENDEDPRICE, L_DISCOUNT))
FROM LINEITEM, ORDERS
WHERE O_ORDERKEY = L_ORDERKEY
GROUP BY L_SHIPDATE, O_SHIPPRIORITY
ORDER BY L_SHIPDATE
```

10GB CCI compressed TPC-H Schema, 2 x CPUs (12 cores), 96GB RAM, SSD storage

	Query without UDF	Query with UDF (no inlining)	Query with UDF (inlining)
Execution time	1.6 seconds	29 minutes 11 seconds	1.6 seconds

Scalar UDF inlining requirements

- Written using the following constructs
  - DECLARE, SET (var declaration/assignments)
  - SELECT (single/multiple var assignments)
  - IF/ELSE (arbitrary nesting levels)
  - RETURN (single or multiple)
  - UDF nested/recursive function calls
  - Relational operations like EXISTS, ISNULL
- No invocation of functions that are
  - time-dependent (GETDATE())
  - has side effects (NEWSEQUENTIALID())
- Uses EXECUTE AS CALLER (default)
- No table variables references
- No table-valued parameters references
- No user-defined types references
- Not natively compiled
  - interop supported
- Not a partition function
- Not referenced in
  - GROUP BY clauses
  - computed columns
  - check constraints
- No signatures added to it

Scalar UDF inlining troubleshoting

- Column **is_inlineable** in **sys.sql_modules**
  - Doesn’t imply it will always be inlined! (e.g. 1000s lines of code)
- Execution Plan
  - If inlined successfully, xml node **<UserDefinedFunction>** will be missing
- Extended Events
  - **tsql_scalar_udf_not_inlineable**

APPROX_COUNT_DISTINCT

- Returns **approximate** number of unique non-null values in groups
  - HyperLogLog algorithm guarantees
    up to 2% error rate within 97% probability
- Fast data exploration with low memory footprint
  - E.g. dashboards, trend analysis, feature selection, etc.
  - Think 10 billion rows,
    1 user using 1,5GB memory vs 100 users using 12MBs
  - Tradeoff: precision, only scenarios where <u>exact values are not necessary</u>!

Demo

- Scalar UDF inlining
- APPROX_COUNT_DISTINCT

Q&A